

Contribution Report

Ensemble Methods: Boosting vs. Bagging
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Fatma Mammadova Ibrahim Mammadov Nihat Rzayev Fidan Allahverdiyeva

Introduction

This report summarizes how work was divided among the four team members for the *Ensemble Methods: Boosting vs. Bagging* project. Each member owned a distinct technical track — Decision Tree, AdaBoost, Random Forest, and Unsupervised/Integration — with effort planned and tracked to stay balanced across the team. The tables below present module ownership, experiments, paper sections, and the overall workload split.

1 Contribution Overview

Member	Primary Modules / Files	Experiments	Paper Sections	Other Tasks	Effort
Fatma Mammadova	<code>src/trees/</code> DecisionTree (CART, Gini/entropy, weighted samples)	Baseline (Exp. 1)	Tree & Baseline	Unit tests, sklearn parity	25%
Ibrahim Mammadov	<code>src/boosting/</code> AdaBoost (SAMME, weight updates)	Head-to-head CV (Exp. 4), Bias–variance (Exp. 6)	AdaBoost derivation, Exp. 4/6	Contribution report coordination	25%
Nihat Rzayev	<code>src/bagging/</code> RandomForest (bootstrap, OOB, parallelism)	RF scaling (Exp. 3), Noise robustness (Exp. 5)	Random Forest	Team-wide test coverage audit	25%
Fidan Allahverdiyeva	<code>src/unsupervised/</code> , preprocessing, <code>run_all.py</code>	Unsupervised analysis (Exp. 7)	Unsupervised, Integration/Conclusion	Slide deck, IEEE formatting	25%

Each track combines one core algorithm implementation, its dedicated experiments, and a matching paper section, so ownership stays traceable end-to-end from code to write-up.

2 Work Distribution

Category	Fatma	Ibrahim	Nihat	Fidan
Implementation	25%	25%	25%	25%
Experiments	25%	25%	25%	25%
Testing / QA	25%	25%	25%	25%
Documentation & Integration	25%	25%	25%	25%
Overall	25%	25%	25%	25%

Workload was estimated per task before implementation began and rebalanced across the four tracks so that no member’s total effort deviates meaningfully from the others.

3 Paper Contribution

Member	Section(s) Authored	Writing Share
Fatma Mammadova	Tree & Baseline	25%
Ibrahim Mammadov	AdaBoost derivation, Exp. 4/6 write-up	25%
Nihat Rzayev	Random Forest	25%
Fidan Allahverdiyeva	Unsupervised, Integration/Conclusion	25%
Joint review, editing & final IEEE formatting		Shared

Every member authored one primary section, and the paper as a whole — introduction, abstract, and conclusion — was reviewed and edited jointly before submission.

Collaboration Statement

All members co-authored the final paper and shared equal writing responsibility. Interfaces between modules (tree, boosting, bagging, data pipeline) were jointly designed and agreed upon. Experimental results were reviewed collectively before submission. Workload across implementation, experiments, testing, and documentation was balanced at approximately 25% per member.